

Algorithms and Data Structures Reference Manual

Chapter 1: Graph Algorithms and Breadth-First Search

1.1 Breadth-First Search (BFS)

Breadth-first search is a fundamental graph traversal algorithm. It systematically explores the edges of a graph to discover every vertex reachable from a starting source vertex s .

BFS uses a First-In, First-Out (FIFO) queue to manage discovered vertices. The algorithm computes the shortest path distance (measured in number of edges) from the source to each reachable vertex.

The running time of breadth-first search is $O(V + E)$, where V is the number of vertices and E is the number of edges in the graph.

Chapter 2: Dynamic Programming Paradigms

2.1 Memoization and Optimal Substructure

Dynamic programming solves optimization problems by combining solutions to overlapping subproblems.

A problem exhibits optimal substructure if an optimal solution to the problem contains within it optimal solutions to subproblems.

For instance, the Bellman-Ford shortest path algorithm employs dynamic programming to find shortest paths in directed graphs with potential negative weights in $O(V * E)$ time.